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Artificial Intelligence
Problems nr 5

Problem 1:

Initial State:

$$e = [3, 2, 4, 1]$$

Heuristic Cost:

The heuristic cost is calculated by counting how many queens are getting killed in the state

First move:

Column 1

$$e_1 = [1, 2, 4, 1] \quad h(e) = 3$$

$$e_2 = [2, 2, 4, 1] \quad h(e) = 3$$

$$e_3 = [4, 2, 4, 1] \quad h(e) = 3$$

Column 2

$$e_4 = [3, 1, 4, 1] \quad h(e) = 2$$

$$e_5 = [3, 3, 4, 1] \quad h(e) = 4$$

$$e_6 = [3, 4, 4, 1] \quad h(e) = 3$$

Column 3

$$e_7 = [3, 2, 1, 1] \quad h(e) = 4$$

$$e_8 = [3, 2, 2, 1] \quad h(e) = 4$$

$$e_9 = [3, 2, 3, 1] \quad h(e) = 3$$

Column 4

$$e_{10} = [3, 2, 4, 2] \quad h(e) = 3$$

$$e_{11} = [3, 2, 4, 3] \quad h(e) = 4$$

$$e_{12} = [3, 2, 4, 4] \quad h(e) = 4$$

Second move:

Column 1

$$e_1 = [1, 1, 4, 1] \quad h(e) = 3$$

$$e_2 = [2, 1, 4, 1] \quad h(e) = 3$$

$$e_3 = [4, 1, 4, 1] \quad h(e) = 4$$

Column 2

$$e_4 = [3, 2, 4, 1] \quad h(e) = 2$$

$$e_5 = [3, 3, 4, 1] \quad h(e) = 3$$

$$e_6 = [3, 4, 4, 1] \quad h(e) = 3$$

Column 3

$$e_7 = [3, 1, 1, 1] \quad h(e) = 4$$

$$e_8 = [3, 1, 2, 1] \quad h(e) = 3$$

$$e_9 = [3, 1, 3, 1] \quad h(e) = 4$$

Column 4

$$\mathbf{e_{10} = [3, 1, 4, 2] \quad h(e) = 0}$$

$$e_{11} = [3, 1, 4, 3] \quad h(e) = 3$$

$$e_{12} = [3, 1, 4, 4] \quad h(e) = 2$$

Answer: $\mathbf{e_{10} = [3, 1, 4, 2] \quad h(e) = 0}$

Problem 2:

- a) Yes, It is possible to solve it using search algorithms
- b) Nodes are segments of space, through which the robot can go and branches are movements
- c) steepest descent or steepest ascent
- d) We could use distance to dashed line (the one in the middle of the room) and distance to obstacle.

$$h = d_{\text{dashed line}} + (d_{\text{max}} - d_{\text{obstacle}})$$

Problem 3:

- a) Yes, with local searching algorithms or Graph search
- b) for graph approach it would be just starting city in the path like [A] and every other city would be mark as unvisited, but for local search it would have to be a starting path that would be later on optimized.
- c) We could define it by using 2-Opt switches or using the graph search it would be all available extensions from the last node excluding already visited ones
- d) In case we would implement graph solution it could be based on number of visited cities multiplied by something plus way back, but in this we just have to look at full distance
- e) Yes, it is because it's able to escape the local optimums and find the global ones, not like the steepest ascent/descent.